

REGEV'S REDUCTION

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$$\text{SIS} \longrightarrow \text{LWE}$$

Reduction between two computational problems: LWE and SIS.

- What are these problems ?
- Where do they come from ?
- What is a reduction ?
- What consequences stem from them ?

POST-QUANTUM CRYPTOGRAPHY

Internet communications are secure thanks to *public-key cryptography*.

↪ Based on computational problems, *e.g.*:

- **FACTO**: given a random $x = pq \in \mathbb{N}$, find p and q .
- **Discrete LOG**: given a random $x \in G = \langle g \rangle$, find $\log_g(x)$.

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Breaking the cryptosystem	$\implies$	solving the problem.
The problem is hard to solve	$\implies$	the cryptosystem is hard to break.

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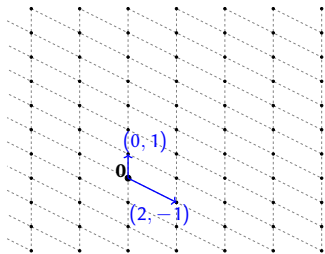
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Shor's algorithm (1994) solves both these problems → need quantum resistant problems.

SIS AND LWE

Lattice generated by $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$: $\mathcal{L} := \{\mathbf{A}\mathbf{u} : \mathbf{u} \in \mathbb{Z}_q^m\}$



Lattice generated by $(0, 1)$ and $(2, -1)$, but also
by $(1, -3)$ and $(1, -4)$

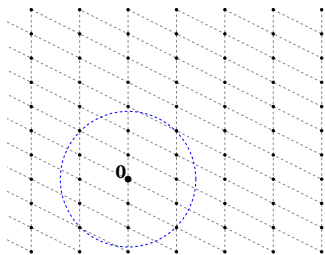
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SHORT INTEGER SOLUTION (SIS)

Given $\mathcal{L}$.

Goal: Find $\mathbf{x} \neq \mathbf{0} \in \mathcal{L}$ s.t. $\|\mathbf{x}\|_2 \leq r$.



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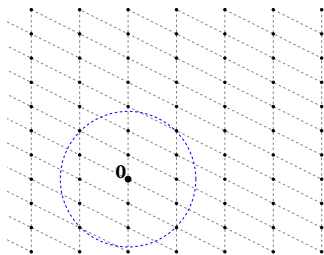
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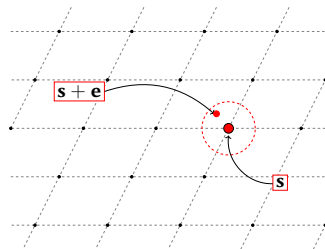


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LEARNING WITH ERRORS (LWE)

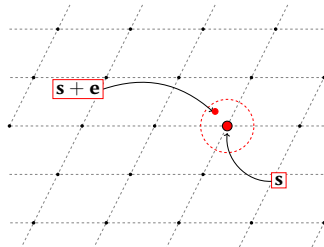
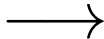
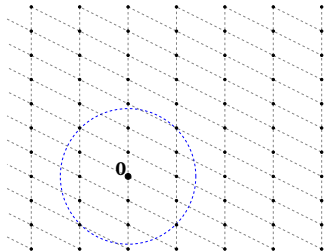
Given $\mathcal{L}$ and $\underbrace{\mathbf{s}}_{\text{secret}} + \underbrace{\mathbf{e}}_{\text{error vector}}$.

Goal: Find $\mathbf{s}$.



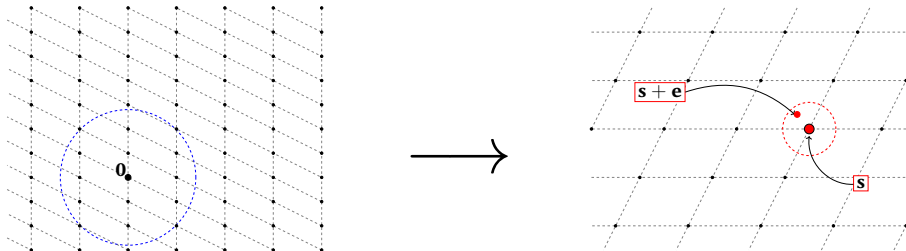
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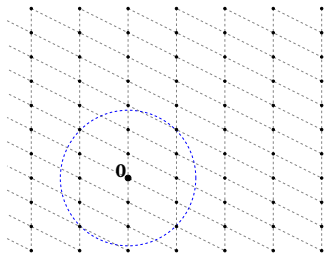
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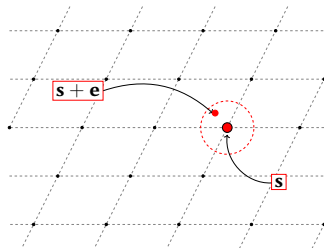
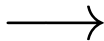
Given an LWE solver (or decoder), one can solve the SIS problem.

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Lattice $\mathcal{L}^* \approx \mathcal{L}^\perp$



Lattice $\mathcal{L}$

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REGEV'S REDUCTION IN A NUTSHELL

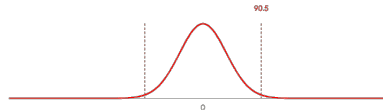
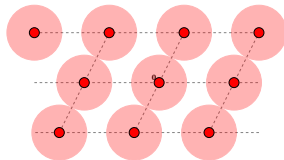
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1. Using the LWE solver, prepare

$$\sum_{\mathbf{x} \in \mathcal{L}} \sum_{\mathbf{e} \in \mathbb{Z}_q^n} f(\mathbf{e}) |\mathbf{x} + \mathbf{e}\rangle.$$



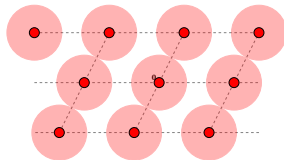
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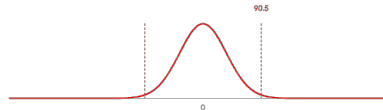
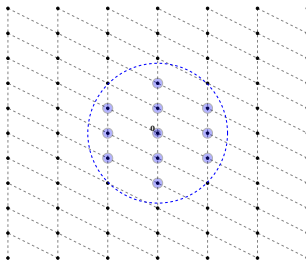
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$$\sum_{\mathbf{y} \in \mathcal{L}^*} \hat{f}(\mathbf{y}) |\mathbf{y}\rangle.$$



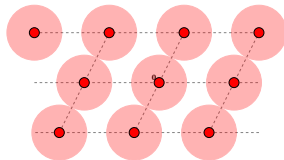
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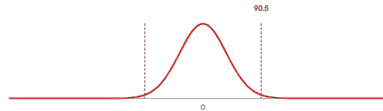
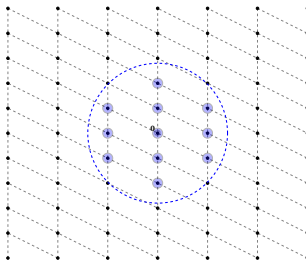
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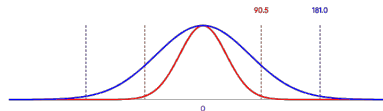


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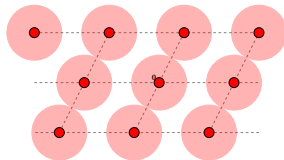
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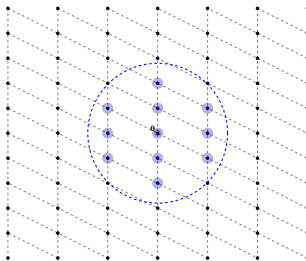
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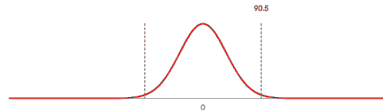


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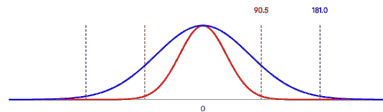
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3. Measure.



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IMPLICATIONS AND RELATED WORKS

Different directions to work on:

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Defined a problem that is hard classically, but easy quantumly, related to a random oracle only.

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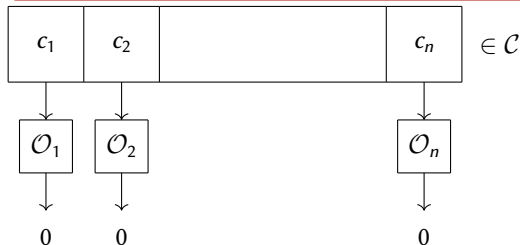
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Given n random oracles $\mathcal{O}_i : \mathbb{F}_q^n \rightarrow \{0, 1\}$.

Find $c \in \mathcal{C}$ s.t. $\mathcal{O}_i(c_i) = 0$ for all $i \in [n]$.

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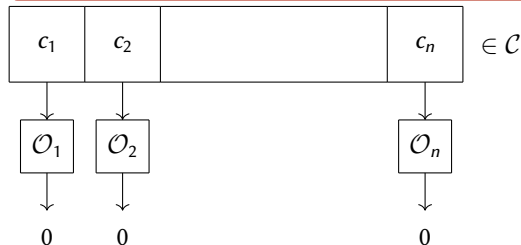
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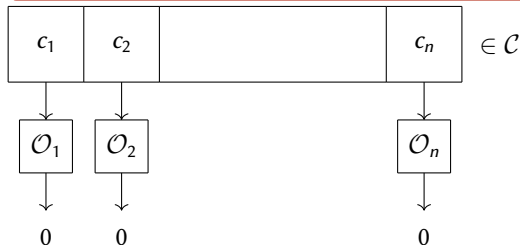
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Easy quantumly:

- “short vector”: $\mathbf{x}$ s.t. $\mathcal{O}_i(\mathbf{x}_i) = 0$ for all $i \longrightarrow \hat{f}(\mathbf{x}) = 1$ if $\mathcal{O}_i(\mathbf{x}_i) = 0$ for all i ;
- choose an appropriate code $\mathcal{C}$ so that f is efficiently decodable.

CONCLUSION

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